

Shouyihuo

An inspectable 3D training simulation

A local-first, bilingual browser prototype connecting observation, diagnosis, simulated treatment and verification to one explicit state model. English is the default.

3 fault cases

3 learning modes

No backend or API key

The screenshot displays the Shouyihuo 3D training lab interface. The top navigation bar includes 'Shouyihuo', '3D training lab', 'Course', 'Workspace', 'Learning records', and language options 'English' and '中文'. The main title is 'Explore the cistern components'. Below the title, a message states: 'Select components to learn their function and observe normal filling and draining. Structure exploration is not scored.' The interface is divided into three main sections:

- Free exploration:** A sidebar on the left with a list of components: 01 Cistern shell, 02 Cistern lid, 03 Supply valve, 04 Supply pipe, 05 Inlet assembly (selected), 06 Float and lever, 07 Flush assembly & seal, and 08 Overflow pipe.
- 3D WORKSPACE:** The central area showing a 3D model of a toilet cistern. It includes a top view, focus selection, and cutaway options. The model shows the internal components like the float valve and flush assembly. A 'Replace lid' button is visible.
- Observe & diagnose:** A panel on the right showing the selected component '05 Inlet assembly'. It includes buttons for 'Record observation', 'Check water level', and 'Check supply'. Below these, there is a 'Model notes' section with text: 'Works with the float to control filling. In normal operation, it stops inflow at the model's working water level.' There is also a 'Before removal' section with a checklist: 'Before removal: supply off → confirm isolation → drain → open lid. Observation and checks need no isolation.' At the bottom, there is a 'Normal operation demo' section with a 'Run retest' button.

At the bottom of the interface, there are two footnotes: 'Teaching prototype. Not reviewed by a repair professional; applies only to the built-in simplified model.' and 'Progress saved automatically · Water level and flow are teaching approximations'.

Actual detailed-model WebGL exploration, recorded on 5 October 2026. Exploration is unscored. English is the default; switching to Chinese preserves the attempt.

01 The problem: keep visual actions and rules consistent

A convincing model is insufficient if an unsafe action silently succeeds or a camera change affects the score. The prototype makes these constraints testable: rejected disassembly preserves the assembly, records the error and prevents a later safety pass.

Built-in course: a simplified gravity-fed toilet cistern with a float-controlled inlet and flapper-style drain seal. Cases cover seal failure, inlet-control failure and a closed supply valve; the last requires no replacement.

Scope: an unaudited teaching model, not repair guidance, a calibrated digital twin or a professional qualification.

One state. Several views.

Typed transitions connect the lesson, 3D scene, assessment and local records.

Typed user events

Observe, diagnose, remove, replace, retest, tick



Pure simulation reducer

Guards + water approximation + verification + score



Derived interface and validated persistence

React / Three.js read state; localStorage restores it

02 Determinism with bounded claims

The same initial state and event sequence produce the same outcome. The UI advances the model by 0.25 simulation seconds per tick; the reducer uses internal steps of at most 0.05 seconds. Water is normalised teaching state, not a measured hydraulic quantity.

03 Safety is a gate, not a score deduction

Evidence, diagnosis, process and verification receive 25, 30, 25 and 20 points. Passing also requires correct diagnosis and treatment, complete verification and no critical safety error. Even a 100-point attempt may fail.

04 Coherent geometry, on-demand drawing

A fixed-pivot lever follows water without another physics engine. Instanced chain links, hollow sleeves and an in-memory hose normal map add detail. The scene requests frames for visible changes, then settles. Double-click focus selects the same action target; display gestures cannot modify an attempt.

05 Persistence preserves both progress and errors

Schema-validated localStorage keeps separate mode sessions and archives each completed attempt once by ID. The original attempt key and schema remain compatible. Locale is a separate preference; changing language cannot repair a fault or erase an error.

Inspect in the repository: `src/domain/engine.ts`, `src/storage/local.ts`, `src/components/TankModel.tsx`, `tests/engine.test.ts` and `e2e/flows.spec.ts`.

Evidence with clear limits

Review the behaviour, the implementation and the recorded checks together.

06 Verification record

The **5 October 2026 model-detail run** passed type checking, **56 unit tests**, **23 browser tests** and a production build. Logs, software-WebGL checks and fresh desktop/mobile captures are in **docs/portfolio/VALIDATION.md**. Earlier results remain dated historical evidence.

Nine invariant tests exercise 3,888 seeded state transitions and 1,944 persisted continuation steps. Browser tests cover all three cases, real mesh selection, fixed-pivot geometry, double-click focus, safety rejection, saved records, language changes, idle rendering and context loss.

07 A failure that changed the implementation

Demand rendering exposed a label-cache bug: a stationary cutaway change could retain stale occlusion. The correction includes geometry changes in cache invalidation and updates world matrices before raycasting. The browser regression toggles the wall and checks actual label visibility.

In one matched idle scene, actual WebGL draw calls changed from 5,180 over approximately 3 seconds to zero. The simulation clock continued advancing. This demonstrates removed idle drawing; it does not measure frame rate, energy use or native-GPU performance.

08 Review in five minutes

Run. Use `npm ci` and `npm run dev` from the project folder. Open `http://127.0.0.1:5173/review/` for the English-default bilingual introduction.

Interact. Follow `docs/portfolio/REVIEWER_GUIDE.md` to complete the supply-valve case, inspect a score report and refresh the saved record.

Challenge. Start a new attempt, request premature removal and inspect the rejected action. Complete the case to see why a numeric score alone cannot pass.

09 Contribution and limitations

AI-assisted development: the project owner supplied the product brief and acceptance constraints. OpenAI Codex contributed substantial implementation, testing, debugging and documentation. The artefact does not imply unaided solo coding; personal contribution claims require their own evidence.

No professional repair review, learner study, physical validation or improved-learning claim is made. Native-GPU performance, every browser/device and physical printing have not been established. Local records are editable and are not formal certificates. The project demonstrates software design and verification practices, not machine learning or computational fluid dynamics.

Reading order: `README.md`, `docs/MODEL_DETAIL_REFINEMENT.md`, the case study and validation report in `docs/portfolio/`, then source and tests. Contribution details: `docs/portfolio/PROVENANCE.md`.